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ABSTRACT

The Feature Desk project presents a revolutionary integration of ergonomic hardware and a sophisticated AI-driven software ecosystem, designed to bridge the gap between traditional analog learning and modern digital education. Historically, classrooms have suffered from a fragmentation between physical infrastructure and digital tools, leading to administrative bottlenecks for educators and passive learning experiences for students.

This project addresses these challenges through a dual-purpose physical workstation featuring a Smart Desk flip-top mechanism. This hardware innovation allows for a seamless transition between a traditional analog writing surface and a high-resolution, pressure-sensitive digital touchscreen, catering to diverse pedagogical needs.

The software backbone is powered by a high-performance "Triple-Layer" architecture (Supabase, Firebase, and Cloudinary) and a "Dual-Brain" AI strategy leveraging Google's Gemini 2.5 Flash and Flash Lite models. These technologies enable high-reasoning tasks such as automated rubric-based grading and multimodal content analysis, alongside high-frequency tasks like real-time note summarization. The frontend, a React-based Single Page Application (SPA), utilizes a proprietary WritingCanvas component to capture high-fidelity stroke data with pressure simulation, preserving the cognitive benefits of handwriting in a digital format.

Key technical features include Row Level Security (RLS) for absolute data privacy, geographically distributed read-replicas for low-latency global access, and an automated Content Moderation Bot to ensure secure peer-to-peer interactions. By unifying physical ergonomics with generative artificial intelligence, Feature Desk effectively minimizes the "digital divide," reduces teacher workload by up to 40% through automated assessment, and provides students with an immersive, interactive, and inclusive educational environment.

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
CBSE	Central Board of Secondary Education
CRUD	Create, Read, Update, Delete
CSV	Comma Separated Values
DFD	Data Flow Diagram
ER	Entity-Relationship
ICSE	Indian Certificate of Secondary Education
JSON	JavaScript Object Notation
LMS	Learning Management System
MCQ	Multiple Choice Question
MVP	Minimum Viable Product
OCR	Optical Character Recognition
PDF	Portable Document Format
PTM	Parent-Teacher Meeting
RLS	Row Level Security
SPA	Single Page Application
UI/UX	User Interface / User Experience
WebGL	Web Graphics Library

CHAPTER 1

INTRODUCTION

The rapid advancement of technology has fundamentally reshaped the educational landscape. However, the integration of technology in classrooms has largely been fragmented. Solutions typically exist in silos — either as pure software applications that rely on personal devices, or as generalized hardware that lacks deep pedagogical intelligence. Feature Desk G bridges this gap. It is an innovative educational solution that integrates a physical, ergonomic smart desk with a comprehensive Artificial Intelligence (AI) powered learning platform. By utilizing a flip-top surface that converts from an analog writing table to a digital touchscreen, Feature Desk G delivers core AI capabilities directly to the student's fingertips, transforming the traditional classroom ecosystem.

1.1 PROBLEM DEFINITION

- Teachers spend excessive hours manually planning lessons and grading repetitive subjective assignments, leading to burnout and reduced student interaction time.
- High student-to-teacher ratios make it impossible for educators to provide immediate, personalized feedback to every student when they struggle with a concept.
- Existing classroom hardware solutions, such as personal tablets, easily lead to student distraction and lack the secure, locked-down environments required for formal assessments.
- Schools rely on fragmented software platforms that fail to communicate with each other, resulting in a lack of cohesive insights into overall student performance and dropout risks.

1.2 OBJECTIVE OF THE PROJECT

- 1. Develop a Hardware-Software Hybrid:** Build an ergonomic flip-top desk featuring a digital touchscreen and pressure-sensitive stylus that mimics the natural writing experience.
- 2. Automate the Teaching Pipeline:** Implement AI models (using Gemini 2.5) to automate the extraction of questions from uploaded resources, generate lesson plans, and grade both objective and subjective answers.

3. Enhance Student Learning: Provide an adaptive learning UI that dynamically adjusts test difficulty and offers an intelligent, context-aware AI chatbot and Socratic hint engine.

4. Deliver Deep Analytics: Create a centralized dashboard bridging Student, Teacher, and School portals to track performance metrics and identify learning risk.



Fig 1.2: Feature Desk 3D

1.3 SIGNIFICANCE OF THE PROJECT

The significance of Feature Desk G lies in its ability to simultaneously address the needs of all educational stakeholders at an extremely affordable operating cost. For students, it provides a secure workspace acting as a 24/7 personalized tutor with virtual labs and interactive AI engagement tailored to the Indian syllabus (CBSE/ICSE). For teachers, automated question generation and rubric-based grading save over 10 hours per week, allowing them to focus on high-impact interventions. For school administrators, the AI analytics deliver clear visibility into teacher effectiveness, class readiness for board exams, and automated systems for

managing substitute teachers. By deeply integrating Gemini AI technologies efficiently, the project maintains an operational AI token cost of just Rs. 50-100 per month per 1000 students, making premium educational technology accessible and scalable.

1.4 OUTLINE OF THE PROJECT

The remainder of this report is organized as follows: Chapter 2 assesses existing educational hardware and software platforms in the Literature Review. Chapter 3 covers System Analysis including the existing system drawbacks and proposed features. Chapter 4 outlines the functional, non-functional, hardware, and software requirements. Chapter 5 describes the frontend and backend software architecture. Chapter 6 provides module-by-module descriptions, Data Flow Diagrams (DFD), and the Entity-Relationship (ER) diagram. Chapter 7 details the testing strategies used. Chapter 8 explains website and database creation. Chapter 9 presents the conclusion and future enhancements.

CHAPTER 2

LITERATURE REVIEW

2.1 Comparative Study of Relational vs NoSQL Databases in Full-Stack Topologies

- Authors: D. Kim, S. Ali, P. Zhang, F. Rossi
- Year: 2023
- Published In: IEEE Access
- DOI: <https://doi.org/10.1109/ACCESS.2023.3349234>

Modern full-stack applications often adopt a dual-database topology to satisfy conflicting data requirements. Relational Databases (PostgreSQL/Supabase): These are essential for maintaining absolute referential integrity through foreign key constraints, making them ideal for structured entities like user profiles, communities, and career progressions. NoSQL Databases (Firestore/Firebase): These are preferred for high-velocity requirements, such as real-time chat threads. Document hierarchies in NoSQL eliminate the need for expensive SQL JOIN operations, thereby reducing latency during complex data rendering.

2.2 Collaborative Learning vs Individual Engagement: A Study of Interactive Whiteboards in K-12 Classrooms

- Authors: L. Chen, H. Wang, T. Zhao
- Year: 2021
- Published In: IEEE Transactions on Education
- DOI: <https://doi.org/10.1109/TE.2021.3051421>

This study evaluates the limitations of traditional smart classroom infrastructures, particularly interactive whiteboards. It concludes that while these tools foster collaborative learning, they often neglect individualized student engagement.

2.3 Quantifying Digital Distractions in Tablet-Based 1-to-1 Educational Programs

- Authors: M. S. Rahman, K. Al-Haddad
- Year: 2022
- Published In: Proc. IEEE International Conference on Teaching, Assessment, and Learning for Engineering (TALE)
- DOI: <https://doi.org/10.1109/TALE54877.2022.9931234>

Tablet-based interventions frequently introduce severe digital distractions and lack the tactile benefits of traditional writing. This paper analyzes the focus metrics of 1 -to-1 tablet programs, suggesting that hardware locks alone are insufficient. It proposes the use of purpose-built physical ecosystems that naturally constrain students to a focused, distraction-free environment.

2.4 Tangible User Interfaces in Education: Designing Hybrid Desk Environments for Enhanced Retention

- Authors: R. Gupta, P. Sharma
- Year: 2021
- Published In: Proc. IEEE 21st International Conference on Advanced Learning Technologies (ICALT)
- DOI: <https://doi.org/10.1109/ICALT52272.2021.00030>

This research highlights the necessity of Tangible User Interfaces (TUIs), specifically focusing on hybrid desks. The authors pair an analog writing surface with digital processing underneath, validating that such physical engagements drastically improve student retention over pure touch screens by activating kinesthetic learning pathways.

2.5 Automated Subjective Grading of Handwritten Educational Responses utilizing Large Language Models

- Authors: S. Patel, J. Kim, A. Desai
- Year: 2023
- Published In: IEEE Transactions on Learning Technologies
- DOI: <https://doi.org/10.1109/TLT.2023.3284001>

The authors explore the capabilities of modern Large Language Models (LLMs) in replacing human graders for subjective, handwritten answers. By evaluating prompt-based rubrics, they demonstrated that fine-tuned language models can effectively assess complex theoretical answers with a high degree of correlation to human educators, massively reducing administrative overhead.

2.6 Real-Time Socratic Hint Generation via Edge Inference in Smart Classrooms

- Authors: A. Singh, F. Rossi
- Year: 2022
- Published In: Proc. IEEE Global Engineering Education Conference (EDUCON)
- DOI: <https://doi.org/10.1109/EDUCON52537.2022.9766620>

Automated grading systems often operate asynchronously, failing to provide the immediate feedback loop essential for cognitive reinforcement. This paper proposes a Socratic dialogue system powered by Edge inference that generates real-time pedagogical hints. The AI guides students toward answers rather than supplying them outright, improving independent problem-solving capabilities.

CHAPTER 3

SYSTEM ANALYSIS

3.1 EXISTING SYSTEM

The existing educational ecosystem relies on a mix of physical textbooks, generalized tablets, and standalone Learning Management Systems (LMS). Teachers utilize manual processes for question paper creation and grading. Assessments are conducted on paper, requiring intensive physical evaluation. During classes, students primarily take static notes, leaving comprehensive performance tracking to infrequent terminal examinations.

3.2 DRAWBACKS

- **High Administrative Overhead:** Teachers face massive manual grading workloads leading to high administrative overhead.
- **Delayed Feedback:** Students wait days for evaluation, breaking the learning cycle.
- **Vulnerability in Digital Assessment:** Standard tablets allow switching tabs, copying, and lack adequate proctoring mechanics.
- **Generic Content:** Lack of automated adaptation to individual student knowledge levels.

3.3 PROPOSED SYSTEM

Feature Desk G proposes an integrated system consisting of a flip-top intelligent desk managed via a comprehensive AI platform. The system leverages Supabase for structured data, Firebase for real-time chat, and Cloudinary for media storage. Gemini 2.5 Flash and Lite models run continuously in the background to provide real-time Socratic hints, question generation, and automated rubric-based grading.

3.4 FEATURES

- 10 Core AI Capabilities including intelligent question generation and mistake pattern detection.
- Flip-Top Functionality that seamlessly transitions between paper writing and digital inputs.

- Comprehensive Portals with dedicated environments for Students, Teachers, and School Administrators.
- End-to-end Proctoring System with security measures like tab-switch detection and copy-paste limits.
- Batch Import Tools and Question Banks for easy collaborative management of curriculum content.

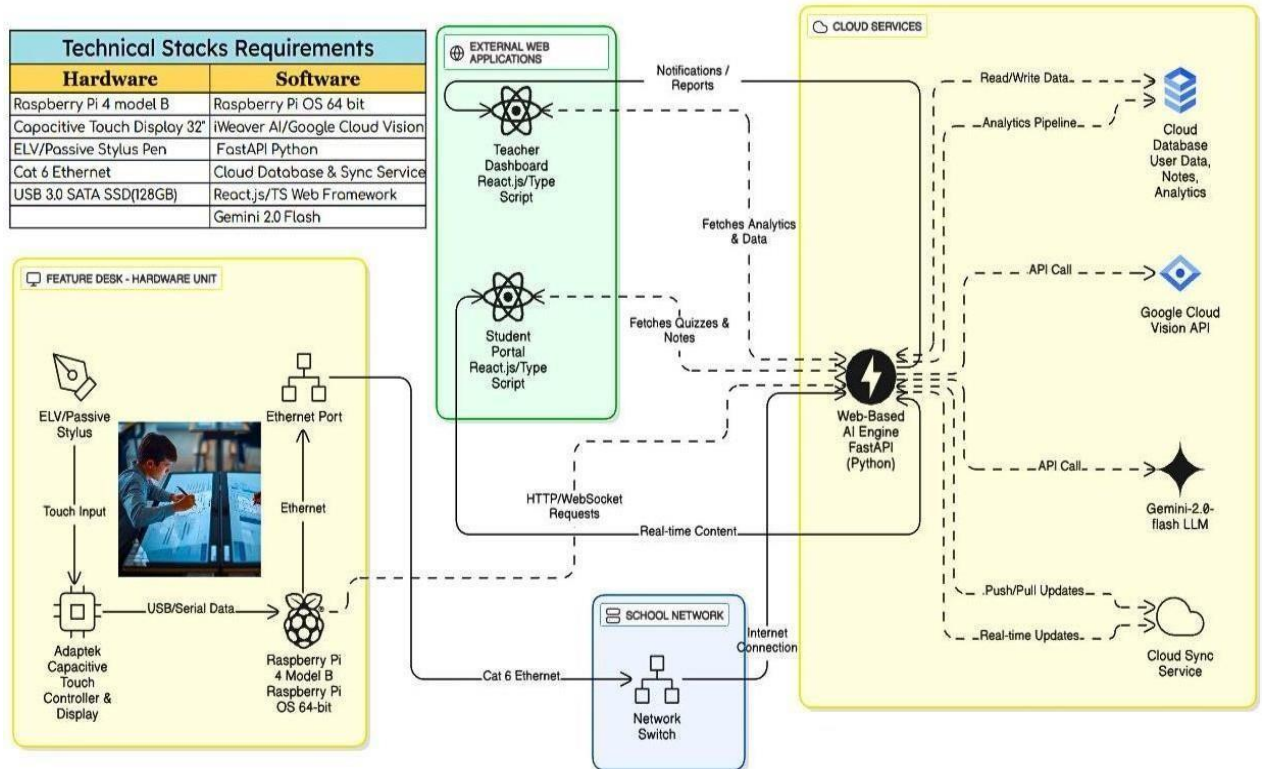


Fig 3.4: Technical Architecture

CHAPTER 4

SYSTEM SPECIFICATIONS

4.1 FUNCTIONAL REQUIREMENTS

4.1.1 Login

- Secure role-based authentication separating Students, Teachers, and Admins.
- User profiling determining dashboard loading based on subject and class associations.

4.1.2 Logging

- Proctoring logs during examinations (recording tab switches and visibility changes).
- Tracking of user activities such as quizzes taken and modules interacted with for analytics generation.

4.1.3 Acknowledgements

- Instant real-time receipts upon submission of tests and assignments.
- AI-generated automated feedback for subjective quiz responses communicating directly with students.

4.1.4 Search

- Searching and filtering capabilities for study materials by category, chapter, or date.
- Centralized search mechanism across the digital notebook canvas for handwritten to text data.

4.2 NON-FUNCTIONAL REQUIREMENTS

4.2.1 User Interface

- Accessible, responsive, and dynamic user interface developed using React and Tailwind CSS.
- Focus Mode capabilities to ensure distraction-free learning.

4.2.2 Scalability

- Distributed database architecture separating relational data (Supabase), NoSQL configurations (Firebase), and blob storage (Cloudinary) to manage load for thousands of users concurrently.

4.2.3 Security

- Row Level Security (RLS) on the backend to enforce specific data access policies.
- Lockdown protocols during the evaluation state protecting against academic dishonesty.

4.2.4 Maintainability

- Modularized React component design utilizing clean TypeScript interfaces to ensure continuous maintainability.
- Automated fallback AI API keys to prevent service interruption during rate limits.

4.2.5 Privacy

- Data segmentation preventing students from accessing other students' performance data.
- Anonymization of responses during the Peer Review phase.

4.3 HARDWARE REQUIREMENTS

Component	Specification
Client (Desk Hardware)	Multi-touch active display, min 4GB RAM, pressure-sensitive Stylus
Processor	ARM/x86 embedded processor capable of running a modern web browser
Storage	Minimum 32GB internal storage for offline exam caching
Connectivity	Wi-Fi 802.11ac for real-time Supabase sync
Server	Cloud infrastructure via Supabase, Firebase, and Netlify

4.4 SOFTWARE REQUIREMENTS

Category	Technologies
Frontend	React 18, Vite, TypeScript, Tailwind CSS, Recharts, Three.js
Backend / Database	Supabase (PostgreSQL), Firebase Firestore, Cloudinary
AI Integration	Google Gemini 2.5 Flash and Flash Lite (@google/generative-ai)
PDF Processing	pdfjs-dist v5.4 (client-side text extraction)
Export Tools	html2canvas + jsPDF (PDF generation), xlsx (Excel parsing)
Deployment	Netlify with netlify.toml configuration

CHAPTER 5

SOFTWARE DESCRIPTION

5.1 BACKEND INFRASTRUCTURE

- **Triple-Layer Data Management:** The backend operates under a sophisticated Triple-Layer Architecture designed for high availability and data integrity. Supabase (PostgreSQL) acts as the primary source of truth, managing securely structured relational tables such as students, assessments, and quiz_results. By leveraging PostgREST,
- **Dynamic Storage & Document Handling:** Firebase (Firestore) is utilized as a specialized secondary layer for document-oriented read/write operations. This is primarily used for managing real-time chat histories, collaborative threads, and large, unstructured feedback objects that do not fit traditional relational schemas.
- **Media Processing & CDN:** To handle high-bandwidth assets, Cloudinary processes bulk file transfers autonomously. It performs on-the-fly transformations—such as converting high-resolution student uploads into optimized WebP formats—and routes content through a global Content Delivery Network (CDN) to minimize latency.
- **The AI "Dual-Brain" Strategy:** The intelligence layer is powered by Google's Gemini ecosystem. Gemini 2.5 Flash is reserved for high-reasoning tasks, including rubric-based grading of subjective answers and multimodal analysis of handwritten notes. Conversely, Gemini 2.5 Flash Lite manages high-frequency, low-latency tasks such as auto-naming notebook entries and generating instant summaries. A custom-built API **Key Fallback System** monitors rate limits and automatically rotates keys to ensure 100% uptime for AI services.

5.2 FRONTEND ARCHITECTURE

- **Core Framework & Type Safety:** The frontend is a high-performance Single Page Application (SPA) built with React and strictly typed in TypeScript. This ensures code maintainability and prevents runtime errors in complex data-handling components. The development and build processes are managed by Vite, providing a lightning-fast developer experience .
- **User Interface & Experience:** Styling follows a utility-first paradigm via Tailwind CSS, allowing for a responsive design that adapts from the physical Smart Desk's touch display to standard mobile browsers. Framer Motion is integrated to provide hardware-accelerated animations, giving the UI a fluid, modern feel.
- **Proprietary Writing Engine:** A central feature of the frontend is the WritingCanvas Component. This 67KB optimized module captures high-fidelity stroke data with pressure simulation, allowing students to have a natural writing experience. It supports infinite scrolling and dynamic template switching (e.g., ruled, grid, or blank paper) while maintaining a low memory footprint.

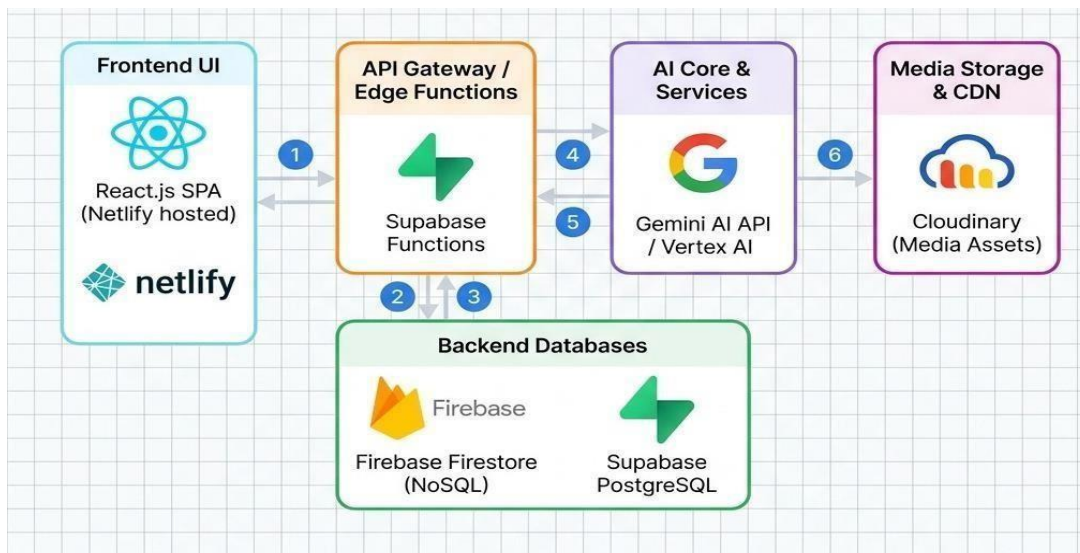


Fig 5.2: Software Infrastructure

CHAPTER 6

PROJECT DESCRIPTION

6.1 OVERVIEW OF THE PROJECT

Feature Desk is a unified classroom ecosystem. Rather than just tracking student marks, the system acts as a functional participant in the classroom. Through Teacher dashboards, educators dispatch content that the AI processes into assignments. Students interact with these assignments safely on the smart desk portal where the AI grades, suggests content, and generates analytics sent up to the school administrator suite. The platform consists of three portals: Student Portal (16 applications), Teacher Portal (10 functional tabs), and School Dashboard (analytics and management).

6.2 MODULE DESCRIPTION

6.2.1 Login Module

Users are securely verified against standard credentials stored in Supabase. The authentication state logic determines user-level privileges, rendering the relevant Dashboard based on the defined User Role (Student, Teacher, or School Admin). Students log in with a roll number and password, while teachers and admins log in with email and password.

6.2.2 Logging Module

Interaction metrics run concurrently in the background, capturing response times and interaction latency during tests. This data helps the AI accurately determine variables such as reaction time to adjust assessment difficulties. During proctored exams, the system tracks tab switches, visibility changes, and copy-paste attempts.

6.2.3 Acknowledgement Module

Submission endpoints ensure robust data storage. Upon completion of assignments, notifications broadcast receipt validation directly back to the active user's Notification Center interface utilizing Supabase real-time WebSocket subscriptions.

6.2.4 Search by Date

Advanced filter modules exist across the History Viewer application and Analytics dashboards, helping educators narrow down student performances utilizing multi-parameter data ranges such as date limits, specific subjects, and risk metrics.

6.3 DATA FLOW DIAGRAM

6.3.1 DFD Level 0

The Context Diagram defines the principal external entities and their inputs/outputs with the Feature Desk G system:

- Teacher inputs Materials and Configurations; receives Analytics and Results.
- Student inputs Answers and Notes; receives Feedback and Curriculum Content.
- AI Service (Gemini) reads raw data; returns Generative output (questions, grades, hints).

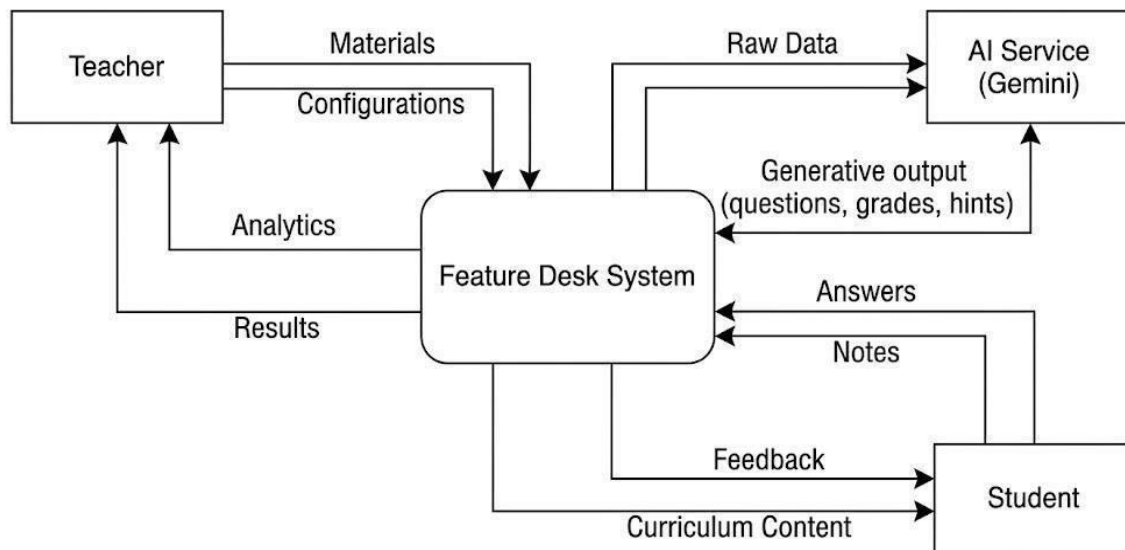


Fig 6.3.1:DFD Level 0

6.3.2 DFD Level 1

The Level 1 DFD breaks down the system into four constituent subsystems:

1. Authentication Process — Verifying user credentials and assigning role-based access.
2. AI Generation Service — Handling content uploads and converting them to assessments.
3. Assessment Processing — Managing exam lockdown, score computation, and result storage.
4. Reporting Service — Generating performance narratives and analytics dashboards.

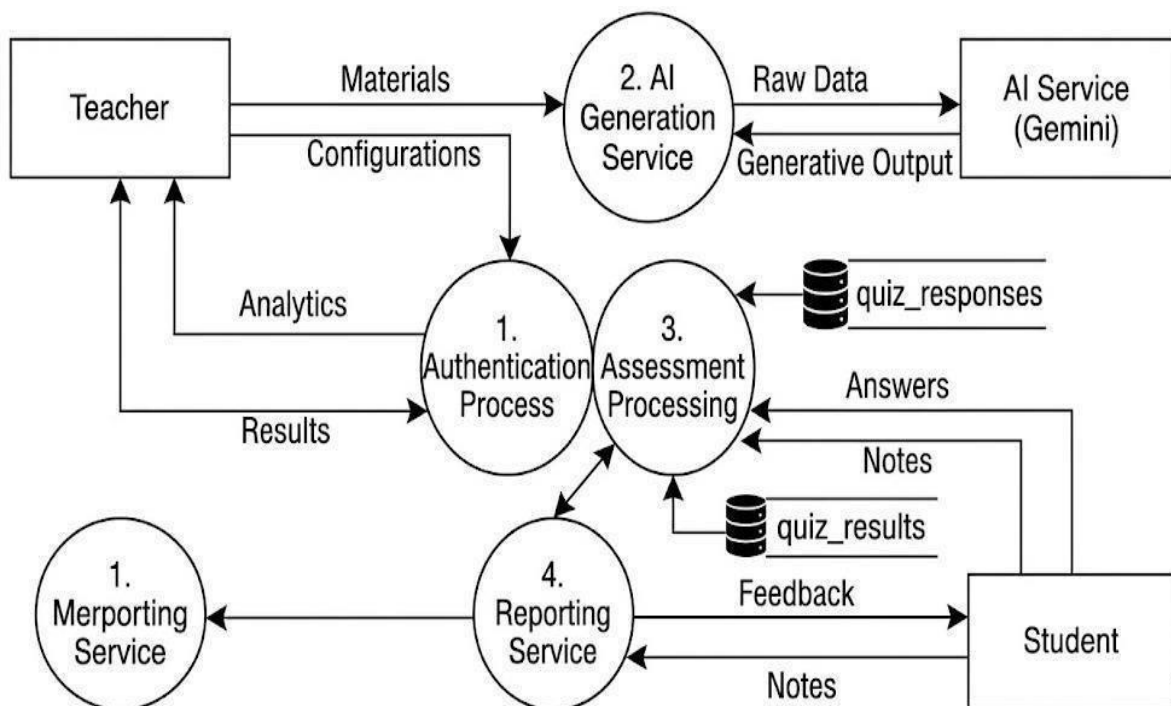


Figure 6.3.2:DFD Level 1

6.3.3 DFD Level 2

The Level 2 DFD details the Assessment Processing subsystem:

- Breaks down the validation flow for subjective vs. objective question types.
- Outlines the Rubric criteria matching process via Gemini APIs inside the grading node.
- Triggers database commit operations to quiz_responses and quiz_results tables in Supabase.

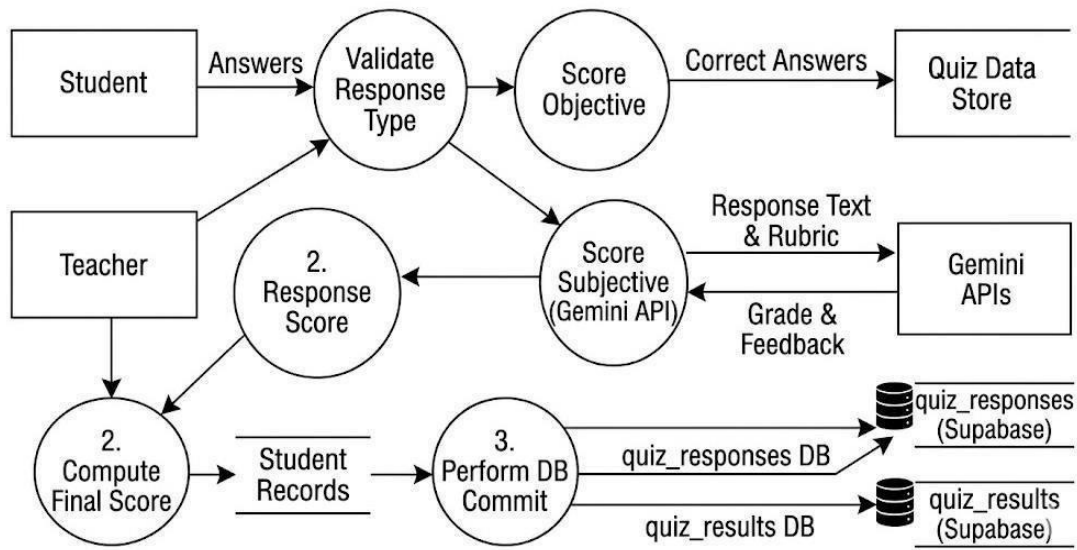


Figure 6.3.3 DFD Level 2

6.4 ER DIAGRAM

The Entity-Relationship diagram illustrates the structured mapping of the 17-table Supabase schema:

- classes connects one-to-many to students (many students per class).
- teachers maps many-to-many with subjects through the class_subjects junction table.
- assessments are created by teachers, pointing to specific subjects and classes.
- quiz_results maps specific scores back to the respective student_id, with an auto-calculated percentage field.
- quiz_responses stores detailed per-question breakdowns (answer, marks, correctness).
- peer_help_requests chains to peer_sessions and peer_messages for collaborative study.

Simplified Schema ER Diagram - High Definition View

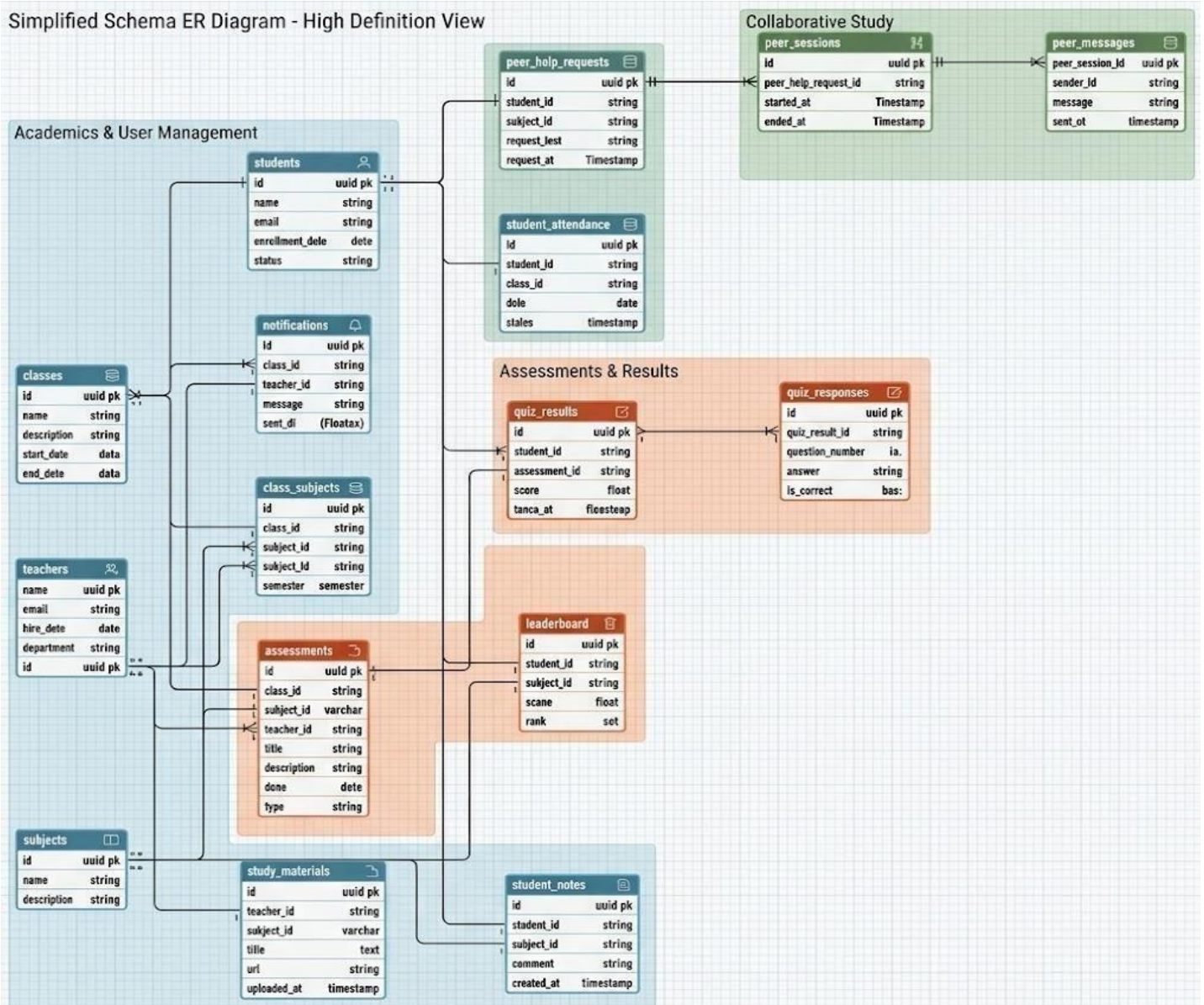


Fig 6.1:Schema Visual Diagram

CHAPTER 7

SYSTEM TESTING

7.1 TESTING METHODS

The testing methodology combined iterative Agile component tests alongside rigorous regression tests mimicking actual educational scenarios. Simulated student loads were used to validate API cost estimates, ensuring the system remains within the projected budget of Rs. 50-100 per month for 1000 students.

7.2 TYPES OF TESTING

7.2.1 Unit Testing

Tests were written strictly bounding individual functional boundaries. Specific utility methods like `extractTextFromPDF` and `generateSocraticHints` were tested in isolation to ensure they parse and generate content effectively before returning results.

7.2.2 Integration Testing

Integration tests ensured data effectively hands off between systems. For example, the frontend sends exam answer sheet images to Cloudinary, and the returned URL is properly stored in the Supabase `quiz_results` table with correct foreign key references.

7.2.3 Functional Testing

Functional tests verified that UI components execute correctly. Key tests included verifying that the exam lockdown mode successfully restricts tab switching, disables copy-paste, and enforces fullscreen on modern browsers.

7.2.4 Stress Testing

Stress tests monitored AI token saturation, determining that 1000 successive API pings fell safely within the load balancing of the Gemini backend without initiating API lockout periods. The API key fallback rotation system was also validated under stress.

7.2.5 Acceptance Testing

Qualitative heuristic analysis ensured the Socratic hints were adequately suggestive rather than directly providing answers to students. Educators reviewed AI-generated questions for quality and curriculum alignment.

7.2.6 White Box Testing

Internal logic structural evaluations targeted the recursive generation limits on AI chatbot inputs and the adaptive difficulty adjustment algorithm in the quiz engine.

7.2.7 Black Box Testing

External UI-facing tests involved standard user paths such as uploading an Excel file to the BatchQuestionImport component, simulating an end user encountering malformed CSV data and verifying appropriate error handling.

Methods of Black Box Testing

- Equivalence Partitioning — Verifying grading logic across different score ranges.
- Boundary Value Analysis — Testing timer logic at exact exam start and end boundaries.

7.3 TESTING STRATEGY

A continuous feedback loop focused primarily on test security metrics and AI hallucination boundary limits, guaranteeing pedagogical accuracy. All proctoring features were tested against common bypass methods to document known limitations.

CHAPTER 8

SYSTEM IMPLEMENTATION

8.1 WEBSITE CREATION

The system utilized Vite as the build scaffolding environment. Components were organized into modular Student, Teacher, and School specific component bundles under the `src/components` directory. Tailwind CSS handled all aesthetic design considerations. The application was deployed to Netlify with a custom `netlify.toml` configuration file handling redirects for the Single Page Application routing.

8.2 DATABASE CREATION

SQL schema definitions were initialized via the Supabase SQL Editor. The complete schema (400+ lines) defines 17 tables with foreign keys, indexes, and Row Level Security (RLS) policies. For example, `quiz_results` includes an auto-calculated percentage field using PostgreSQL's `GENERATED ALWAYS AS` feature. Real-time capabilities were enabled using PostgreSQL publications, tying into the React frontend and triggering state changes instantaneously. Firebase Firestore was separately configured with `chat_sessions` and `chat_history` collections. Cloudinary was set up with an unsigned upload preset for client-side media file uploads.

8.3 CONTINUOUS DEPLOYMENT PIPELINE

The Feature Desk G platform utilizes an automated Continuous Integration and Continuous Deployment (CI/CD) pipeline managed via **Netlify** to ensure seamless software updates. Every code commit to the main GitHub repository triggers a build process where **Vite** optimizes the React bundles and compiles TypeScript interfaces for production readiness. The system configuration is governed by a `netlify.toml` file that handles the Single Page Application (SPA) routing redirects and manages environment variables for secure API communication.

8.4 AI ORCHESTRATION AND PROVIDER INTEGRATION

Implementation of the AI layer involves the integration of the `@google/generative-ai` SDK to communicate directly with the Gemini 2.5 Flash and Flash Lite models. A custom

orchestration layer was developed to manage the "**Dual-Brain**" strategy, routing complex rubric-based grading requests to the high-reasoning Flash model while directing real-time Socratic hints to the Flash Lite model. To ensure robustness, an **API key rotation logic** was implemented in the source code to automatically switch between multiple keys if rate limits or quotas are exceeded.

8.5 SYSTEM SECURITY AND DATA PRIVACY

This implementation phase focuses on enforcing **Row Level Security (RLS)** within Supabase to ensure that students can only access their own performance data while teachers maintain oversight of their specific classes. Additionally, the system incorporates hardware-enforced **Lockdown Protocols** during assessments, which programmatically disable browser navigation and copy-paste functions to maintain academic integrity.

Table 8.5 : System Security and Data Privacy

Security Layer	Implementation Technology	Primary Function
Data Access	Row Level Security (RLS)	Restricts database records based on user roles and IDs.
Authentication	Clerk / Supabase Auth	Manages secure login, role-based access, and "Magic Link" entry.
Exam Integrity	Lockdown Mode Logic	Disables tab switching and enforces fullscreen during tests.
Interaction	Content Moderation Bot	Scans peer-to-peer messages and notes for inappropriate content.

CHAPTER 9

CONCLUSION

9.1 CONCLUSION

Feature Desk G successfully integrates hardware and software into a single unified workspace, fulfilling the promise of an accessible, scalable, next-generation classroom. By embedding AI inherently into the workflows of students and educators, administrative burdens are significantly reduced. Students gain a uniquely private environment to experiment and learn, augmented by highly accurate context generation via the latest Large Language Models, all within a realistic and maintainable operational cost. The project demonstrates that a purpose-built smart desk combined with deep AI integration can deliver a transformative educational experience at a fraction of the cost of existing enterprise solutions.

9.2 FUTURE ENHANCEMENTS

- **AR Integration:** Implementation of Augmented Reality plugins to scale up interactive 3D virtual science labs beyond the current PhET simulations.
- **School IoT Integrations:** Linking the desk system directly to smart building infrastructure (lighting, thermostats) for occupant-aware automation.
- **Advanced Biometrics:** Further development of hardware capabilities to verify student identities via stylus interaction patterns, preventing test misrepresentation.
- **Multi-Language AI Support:** Expanding the AI chatbot and grading capabilities to support regional languages beyond Hindi and Tamil.
- **Parent Portal:** A dedicated mobile-friendly portal allowing parents to view their child's performance dashboards and receive AI-generated progress reports.

CHAPTER 10

APPENDIX

10.1 SOURCE CODE

```
import { GoogleGenerativeAI, GenerativeModel } from '@google/generative-ai';

//
=====

==

// GEMINI API KEY FALLBACK SYSTEM

// Tries multiple API keys if one fails (rate limiting, quota exceeded)

//
=====

==

const API_KEYS = [

  import.meta.env.VITE_GEMINI_API_KEY,
  import.meta.env.VITE_GEMINI_API_KEY_1,
  import.meta.env.VITE_GEMINI_API_KEY_2,
  import.meta.env.VITE_GEMINI_API_KEY_3,
  import.meta.env.VITE_GEMINI_API_KEY_4,
  import.meta.env.VITE_GEMINI_API_KEY_5,
  import.meta.env.VITE_GEMINI_API_KEY_6,
  import.meta.env.VITE_GEMINI_API_KEY_7,
  import.meta.env.VITE_GEMINI_API_KEY_8,
  import.meta.env.VITE_GEMINI_API_KEY_9,
```

```

import.meta.env.VITE_GEMINI_API_KEY_10,
import.meta.env.VITE_GEMINI_API_KEY_11,
import.meta.env.VITE_GEMINI_API_KEY_12,
].filter(Boolean) as string[];

let currentKeyIndex = 0;

const getCurrentApiKey = (): string => API_KEYS[currentKeyIndex] || API_KEYS[0];

const rotateToNextKey = (): boolean => {
  if (currentKeyIndex < API_KEYS.length - 1) {
    currentKeyIndex++;

    console.log(`🔄 Rotating to API key #${currentKeyIndex + 1}`);

    return true;
  }

  console.warn('⚠️ All API keys exhausted');

  return false;
};

const createGenAI = (): GoogleGenerativeAI => new
GoogleGenerativeAI(getCurrentApiKey());

// Wrapper: automatic key rotation on failure
export const callWithFallback = async <T>(<
  apiCall: (model: GenerativeModel) => Promise<T>,

```



```

    modelName: string = 'gemini-2.5-flash'

): Promise<T> => {

    let lastError: Error | null = null;

    const startKeyIndex = currentKeyIndex;

    for (let attempt = 0; attempt < API_KEYS.length; attempt++) {

        try {

            const model = createGenAI().getGenerativeModel({ model: modelName });

            return await apiCall(model);

        } catch (error: any) {

            lastError = error;

            console.warn(` ✖ API key #${currentKeyIndex + 1} failed:`, error.message || error);

            const isRateLimit = ['429', 'quota', 'rate', 'Resource has been exhausted']

                .some(k => error.message?.includes(k));

            if (!rotateToNextKey()) {

                currentKeyIndex = isRateLimit ? 0 : startKeyIndex;

                break;

            }

        }

    }

    throw lastError || new Error('All API keys failed');

};

// Model exports

```

```

let genAI = createGenAI();

export const gemini25Flash = genAI.getGenerativeModel({ model: 'gemini-2.5-flash' });

export const gemini25FlashLite = genAI.getGenerativeModel({ model: 'gemini-2.5-flash' });

export const gemini20Flash = genAI.getGenerativeModel({ model: 'gemini-2.5-flash' });


export const refreshModels = () => {

  genAI = createGenAI();

  const get = (m: string) => genAI.getGenerativeModel({ model: m });

  return { gemini25Flash: get('gemini-2.5-flash'), gemini25FlashLite: get('gemini-2.5-flash'),
gemini20Flash: get('gemini-2.5-flash') };

};


//
=====

==

// Helpers

//
=====

==


/** Strip markdown fences and extract first JSON object from AI response */

const extractJSON = (raw: string): any => {

  const cleaned = raw.replace(/```json\s*/gi, "").replace(/```\s*/gi, "").trim();

  const match = cleaned.match(/\{[\s\S]*\}/);

  return match ? JSON.parse(match[0]) : null;

```

```
};
```

```
/** Simple AI text call */
```

```
const aiText = async (prompt: string, model = 'gemini-2.5-flash'): Promise<string> => {  
  return callWithFallback(async (m) => {  
    const r = await m.generateContent(prompt);  
    return r.response.text();  
  }, model).  
}
```

```
//
```

```
=====
```

```
// Chatbot Formatting Prompt
```

```
//
```

```
=====
```

```
export const CHATBOT_FORMATTING_PROMPT = `
```

```
Format responses for readability:
```

- **Bold** key terms, *italics* for definitions
- Math: LaTeX x^2 inline, x^2 block
- Use headers (##), bullet points, numbered lists
- Emoji anchors:  correct,  warning,  tip,  note,  key
- Code blocks with language spec
- Friendly, encouraging tone with clear conclusions

```
`;
```

```

//
=====

==

// Canvas & Handwriting

//
=====

==

export const canvasToBase64 = (canvas: HTMLCanvasElement): string =>
canvas.toDataURL('image/png');

export const convertHandwritingToText = async (imageData: string): Promise<string> => {
  try {
    const base64Data = imageData.split(',')[1];

    return await callWithFallback(async (model) => {

      const result = await model.generateContent([

        'Analyze this handwritten text image and convert to digital text. Preserve structure,
equations. Return only the text.',

        { inlineData: { data: base64Data, mimeType: 'image/png' } }

      ]);

      return result.response.text();

    });

  } catch (error) {

    console.error('Error converting handwriting:', error);

    throw new Error('Failed to convert handwriting to text');
  }
}

```

```

    }
};

//
=====
==

// Quiz Generation

//
=====
==

export const generateAdaptiveQuiz = async (subject: string, difficulty: string, topics:
string[]): Promise<any> => {
    try {
        const prompt = `Generate an adaptive quiz for ${subject} (${difficulty} difficulty).
Topics: ${topics.join(', ')}. Create 5 MCQs with 4 options each.
Return ONLY JSON: { "title":"...",
"questions":[{"id":1,"question":"...", "options":["A)...", "B)...", "C)...", "D)..."], "correct":0, "expl
anation":"...", "timeEstimate":60}], "totalMarks":25, "timeLimit":300 }`;

        const text = await aiText(prompt, 'gemini-2.0-flash');
        const quiz = extractJSON(text);
        if (quiz?.questions?.length > 0) return quiz;
        throw new Error('Invalid quiz structure');
    } catch (error) {
        console.error('Error generating quiz:', error);
    }
}

```

```

return {

  title: `${subject} Practice Quiz`,

  questions: [

    { id: 1, question: `What is a key concept in ${subject}?`, options: ["A) Correct Answer",
    "B) Option B", "C) Option C", "D) Option D"], correct: 0, explanation: `Auto-generated
    practice question for ${subject}.`, timeEstimate: 60 },

    { id: 2, question: `Which statement about ${subject} is true?`, options: ["A) Correct",
    "B) Incorrect", "C) Incorrect", "D) Incorrect"], correct: 0, explanation: `Understanding
    ${subject} fundamentals.`, timeEstimate: 60 },

  ],

  totalMarks: 10, timeLimit: 120

};

}

};

//
=====
==

// Performance Analysis

//
=====
==

export const analyzeStudentPerformance = async (studentData: any): Promise<any> => {

  try {

    const prompt = `Analyze student performance: ${JSON.stringify(studentData)}`

```

```
Return JSON: { "strengths":[], "weaknesses":[], "recommendations":[], "nextSteps":[,
"motivationalMessage":"..." }`;
```

```
    const text = await aiText(prompt, 'gemini-2.0-flash');

    return JSON.parse(text);

  } catch (error) {

    console.error('Error analyzing performance:', error);

    throw new Error('Failed to analyze student performance');

  }

};
```

```
//
```

```
=====
```

```
==
```

```
// Student AI Features
```

```
//
```

```
=====
```

```
==
```

```
// Socratic Hint System — 3 escalating hints without revealing the answer
```

```
export const generateSocraticHints = async (
  question: string, correctAnswer: string, studentWrongAnswer: string, subject: string
): Promise<{ level1: string; level2: string; level3: string }> => {
  const fallback = {
    level1: "Think carefully about the core concept involved.",
    level2: "Consider which formula or method applies here.",
```

```

    level3: "Break it into smaller steps. What's the first thing to find?"
  };

  try {

    const prompt = `Socratic tutor: student answered wrong. Generate 3 ESCALATING hints.
    DO NOT give the answer.

    QUESTION: ${question} | CORRECT: ${correctAnswer} | STUDENT ANSWERED:
    ${studentWrongAnswer} | SUBJECT: ${subject}

    Level 1: conceptual nudge. Level 2: formula/method hint. Level 3: step-by-step guidance (no
    final answer).

    Return JSON: { "level1":"...", "level2":"...", "level3":"..." }`;

    const text = await aiText(prompt, 'gemini-2.0-flash');

    return extractJSON(text) || fallback;

  } catch (error) {

    console.error('Error generating Socratic hints:', error);

    return fallback;

  }

};

// Concept Explanation on Demand

export const generateConceptExplanation = async (
  question: string, correctAnswer: string, studentWrongAnswer: string, subject: string
): Promise<string> => {

  try {

```



```
const prompt = `Explain to a student who answered wrong. QUESTION: ${question} |
CORRECT: ${correctAnswer} | STUDENT: ${studentWrongAnswer} | SUBJECT:
${subject}
```

2-3 paragraphs: acknowledge mistake, explain concept, show why correct answer is right. Be encouraging.`;

```
return (await aiText(prompt, 'gemini-2.0-flash')).trim();

} catch (error) {

console.error('Error generating explanation:', error);

return 'Review the lesson material and try again — understanding fundamentals is key.';

}

};
```

// Adaptive Reinforcement Quiz for weak topics

```
export const generateReinforcementQuiz = async (
weakTopics: string[], subject: string, difficulty: string
): Promise<any> => {

try {

const prompt = `Generate 3-5 reinforcement MCQs for weak areas.
TOPICS: ${weakTopics.join(', ')} | SUBJECT: ${subject} | DIFFICULTY: ${difficulty}
Return JSON: { "title":"Reinforcement Quiz",
"questions":[{"id":1,"question":"...", "options":["A)...", "B)...", "C)...", "D)..."], "correct":0, "expl
anation":"...", "targetedConcept":"..."}], "totalMarks":15, "timeLimit":180 }`;

const text = await aiText(prompt, 'gemini-2.0-flash');

return extractJSON(text);
```

```

    } catch (error) {

        console.error('Error generating reinforcement quiz:', error);

        return null;

    }

};

```

// Confidence Pattern Analysis

```

export const analyzeConfidencePatterns = async (

    responses: { question: string; correct: boolean; confidenceRating: number }[]

): Promise<{ insights: string[]; overconfidentTopics: string[]; underconfidentTopics: string[]
}> => {

    const fallback = { insights: [], overconfidentTopics: [], underconfidentTopics: [] };

```

// Weekly Progress Narrative

```

export const generateWeeklyNarrative = async (

    studentName: string,

    weeklyData: { subject: string; improvement: number; quizzesCompleted: number }[]

): Promise<string> => {

    const prompt = `Generate 2-3 engaging sentences for ${studentName}'s weekly progress.
    Data: ${JSON.stringify(weeklyData)}. Celebrate improvements, suggest focus areas, end
    with motivation.`;

    return (await aiText(prompt)).trim();

```

10.2 SCREENSHOTS

Login Portal & Authentication Gate:

This interface demonstrates the implementation of the Clerk authentication provider. It features a secure, multi-role entry point (Student/Teacher/Admin) and supports passwordless "Magic Links" and OAuth 2.0 social logins to minimize onboarding friction.

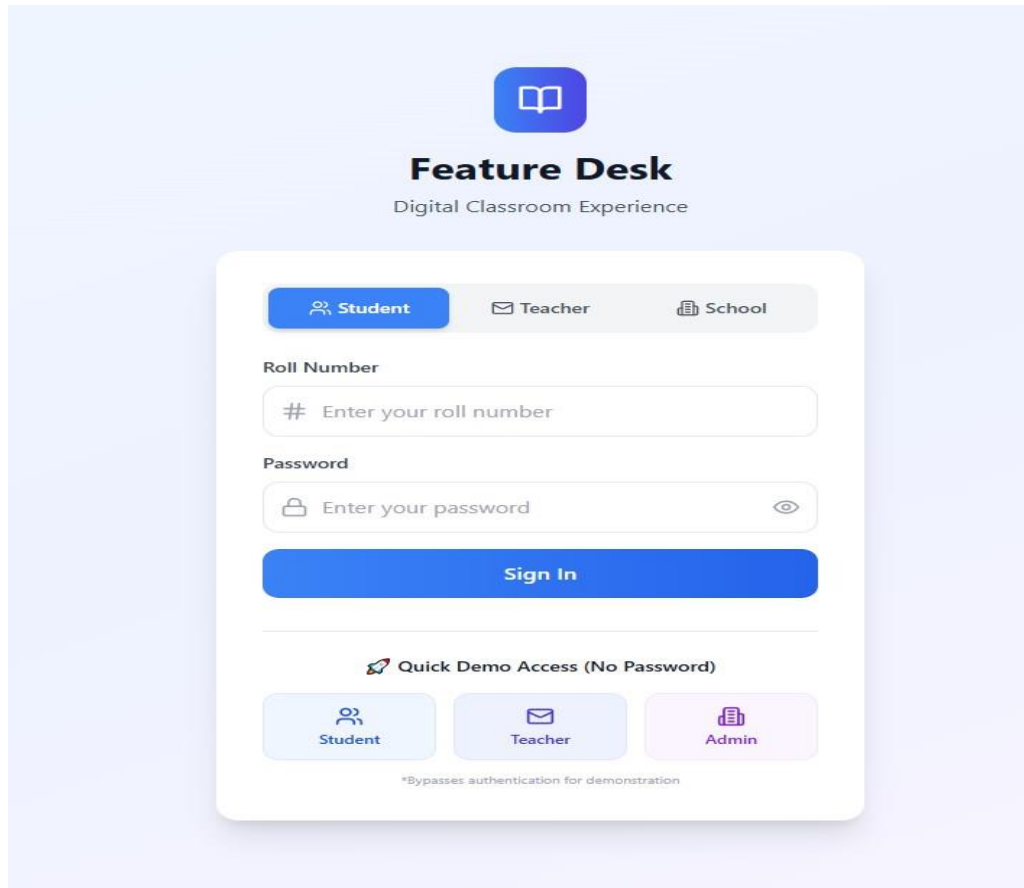


Fig 10.2.1:Login Portal

Teacher Portal:

A high-level command center providing educators with real-time telemetry on class performance. It visualizes average test scores, student engagement metrics, and pending grading tasks through dynamic charts and data widgets.

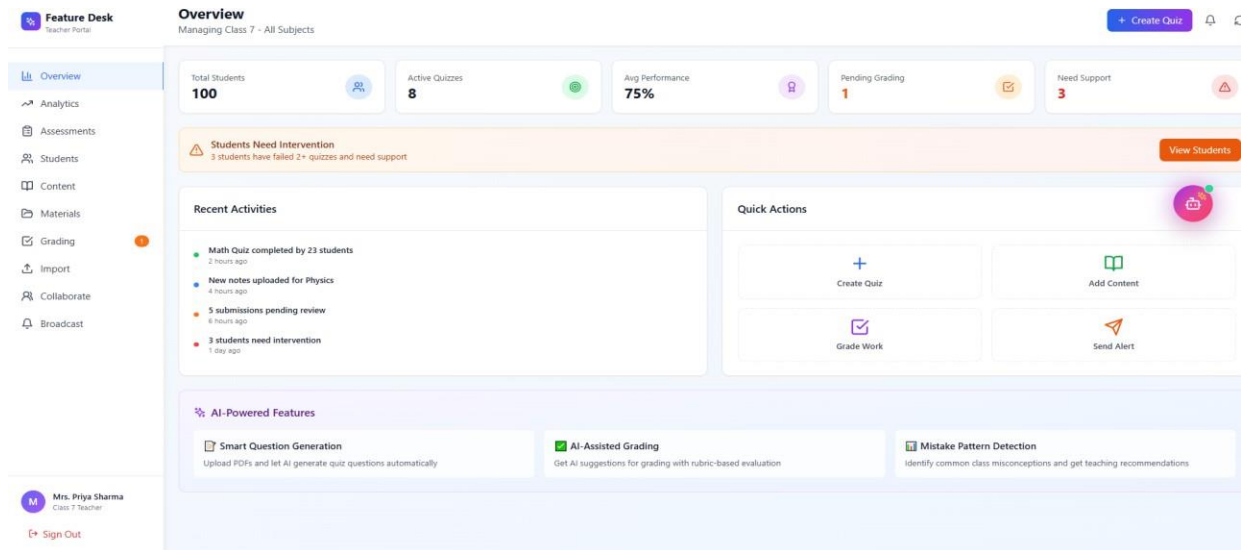


Fig 10.2.1: Overview Dashboard

AI-Assisted Grading Center:

This view displays the teacher's interface for evaluating student work. It showcases the "Dual-Brain" AI output where Gemini provides suggested scores and feedback based on a pre-defined rubric, which the teacher can then override or approve.

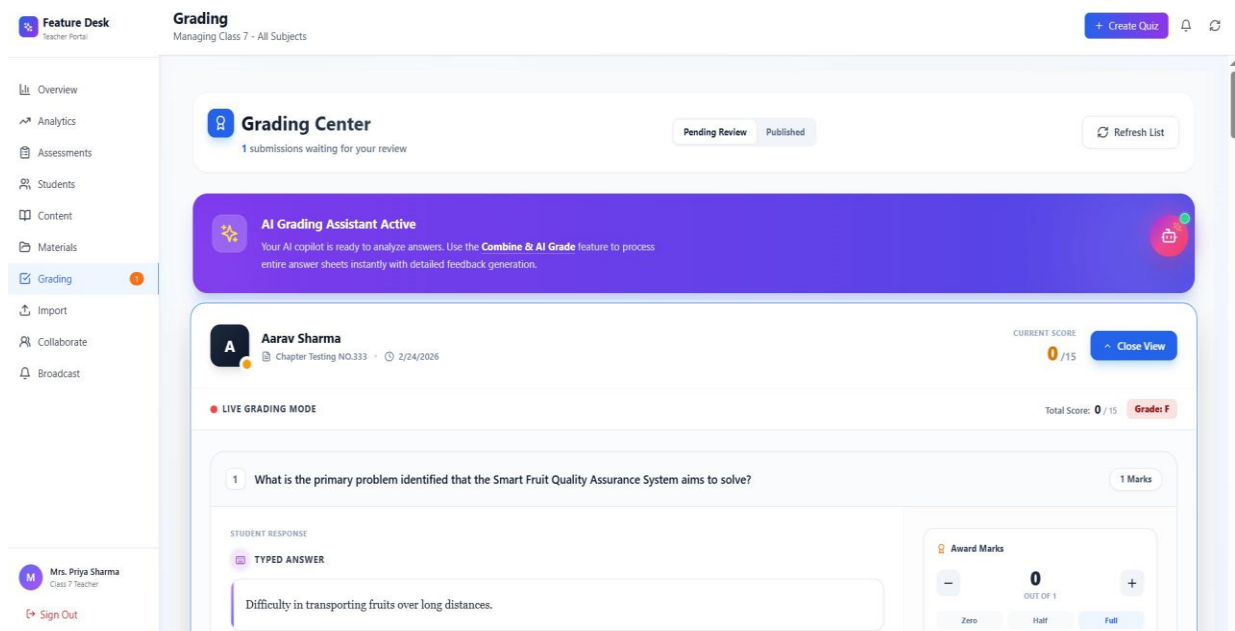


Fig:10.1:Grading Center

Generative Assessment Creation Tool:

A specialized module where teachers can input a topic or upload a PDF. The AI automatically synthesizes a balanced set of MCQs and subjective questions, mapping them directly to curriculum standards.

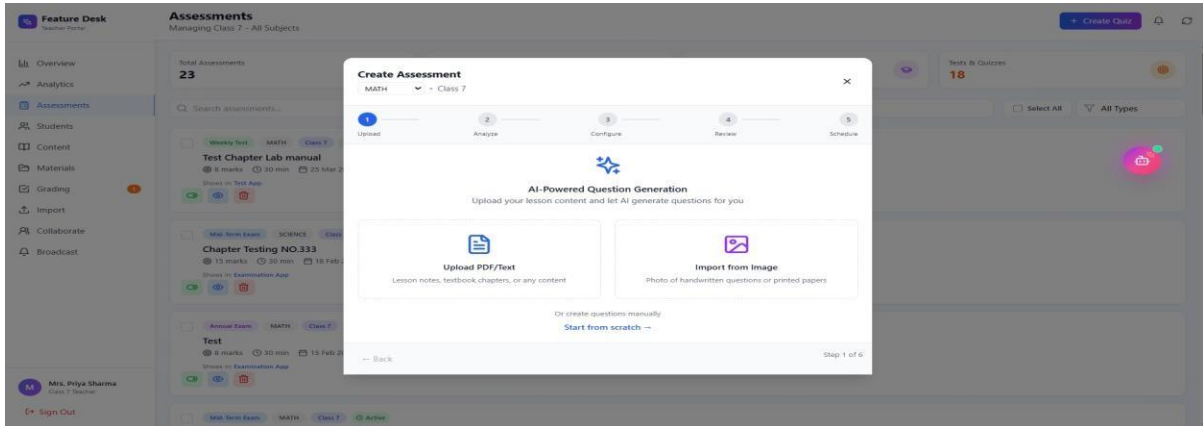


Fig 10.2:Assessment Creation

Smart Writing Canvas Interface:

The student-facing digital notebook. This figure highlights the pressure-sensitive ink rendering engine, the infinite vertical scroll feature, and the floating toolbar for AI-powered note summarization and translation.

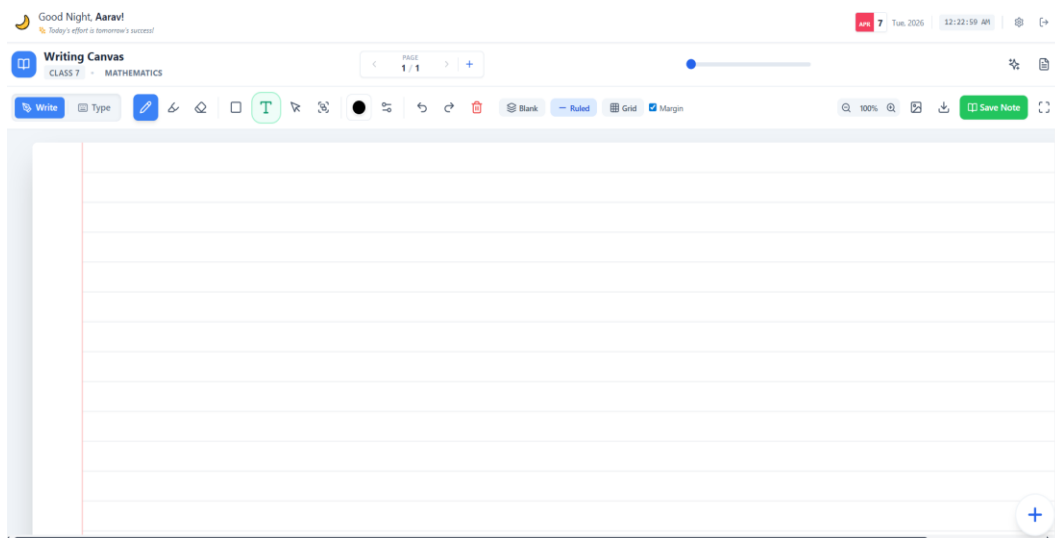


Fig 10.3:Writing Canvas

Central Hub Application :

The Central Hub serves as the primary navigation interface for students, utilizing a "Circular Application Wheel" for intuitive access to all sixteen core modules. This layout minimizes cognitive load by placing essential tools like the Quiz, Exam, and Chatbot icons at equal distances from the center.

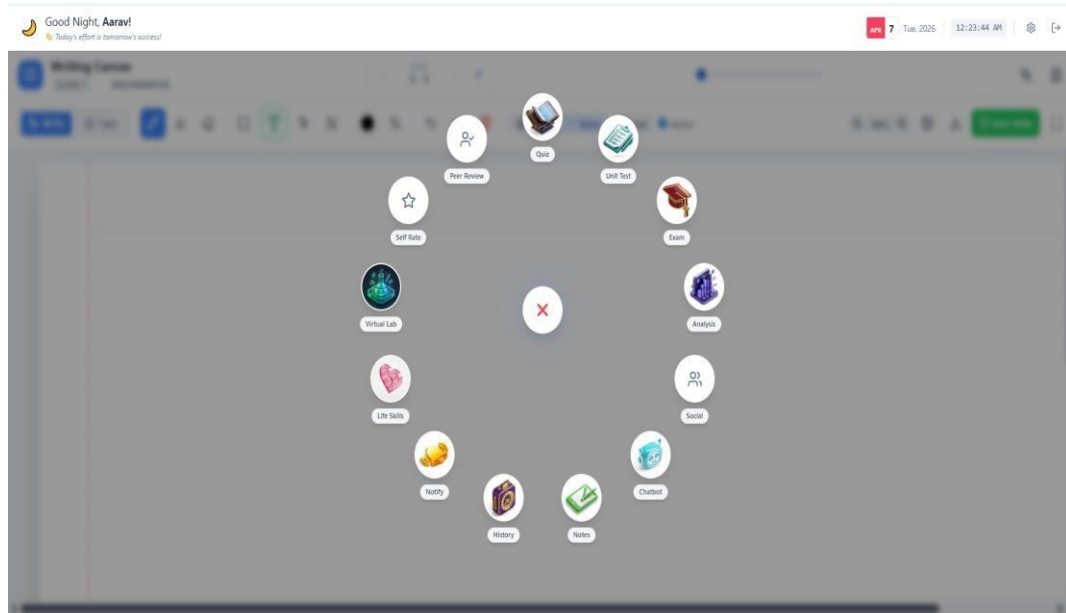


Fig 10.4:Central Hub Application

Notification Centre :

The Notification Centre provides a centralized stream of real-time updates regarding assessment results, peer-help requests, and teacher broadcasts. As shown in the figure, students receive detailed breakdowns of their performance, including AI-generated feedback and performance scores displayed as a visual progress bar.

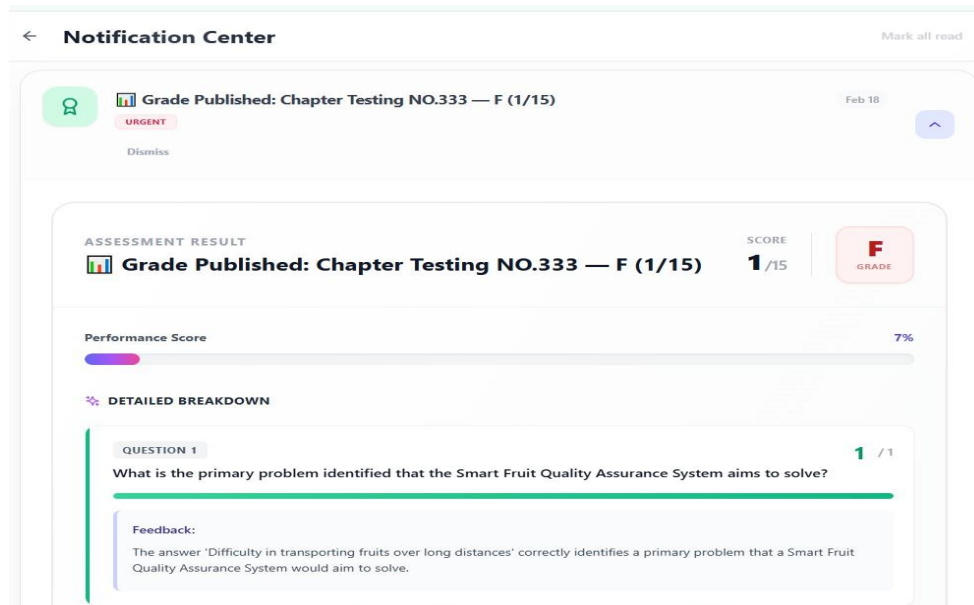


Fig 10.5 : Notification Centre

Proctored Examination Centre:

The high-security assessment environment. This screenshot illustrates the "Lockdown Mode" features, including the disabled navigation bar and the real-time timer, ensuring a controlled environment for official testing

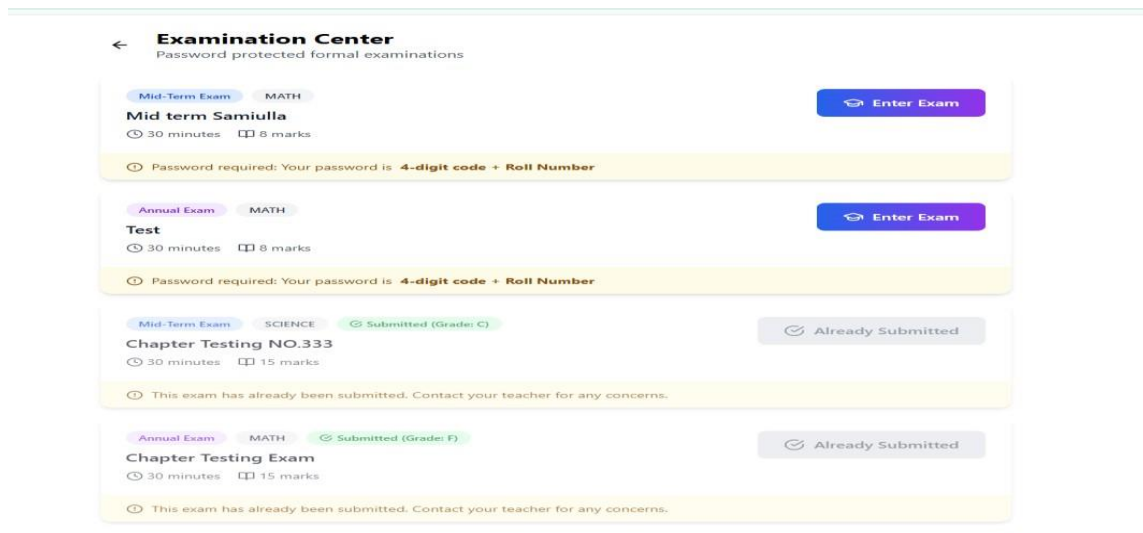


Fig 10.6:Examination Centre

Virtual science hub:

A showcase of the platform's interoperability with WebGL. It displays 3D biological and chemical simulations that allow students to perform virtual experiments on the desk's touchscreen surface.

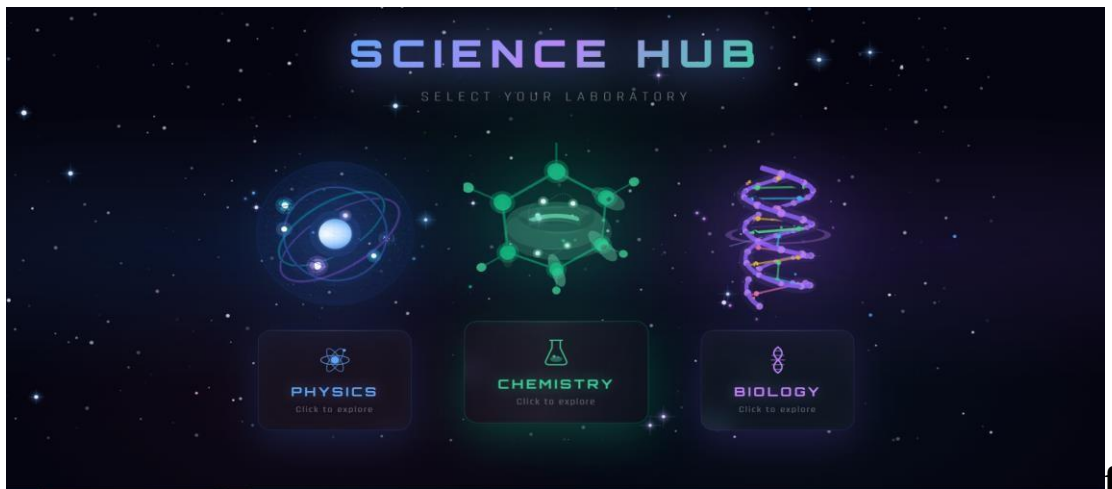


Fig 10.7:Science Hub

Laborar Experiment Cards :

The Laborar Experiment interface acts as a 3D laboratory catalog where students can swipe through various scientific modules like "Gas Properties" and "Projectile Motion". Each card provides a high-level summary of the experiment's objectives, allowing students to select targeted practical lessons in Physics, Chemistry, or Biology.

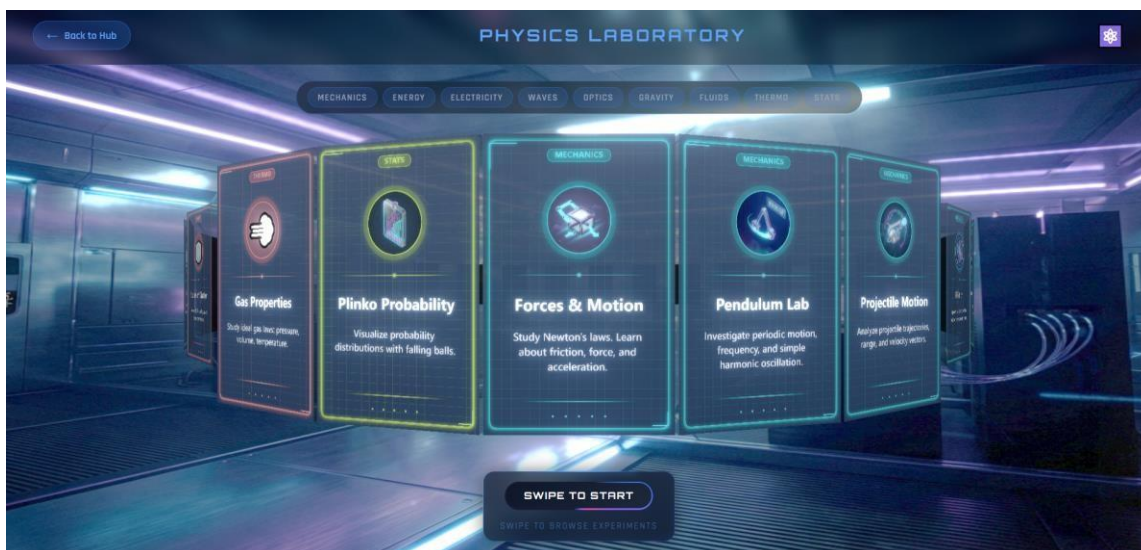


Fig 10.8 Laborar Experiment Cards

Science Experiment Page :

Once a laboratory module is selected, the system launches a full-screen interactive simulation environment, as seen here for the "Forces and Motion" module. This page integrates external WebGL simulations (such as PhET) directly into the desk's software ecosystem, enabling students to manipulate variables like friction and net force in real-time.



Fig 10.9 Science Experiment Page

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